

Problem 4

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Problem 1 Locate the critical points and use the second derivative test to determine whether they correspond to local maxima or local minima. **EX-PLAIN.**

$f(x) = (x + c)^4$ where c is a positive constant

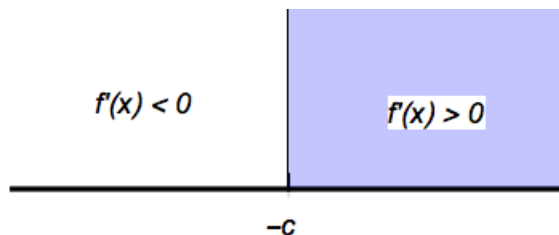
Explanation.

$$f'(x) = 4(x + c)^3, f''(x) = 12(x + c)^2$$

To find the critical points: $f'(x) = 0 = 4(x + c)^3$. This occurs when $x = -c$

Using the second derivative test: $f''(-c) = 12(-c + c)^2 = 0$. The second derivative test was inconclusive so we need to use the first derivative test.

Making a sign chart we see that if we plug something a little less than $-c$ into f' , we'll get a negative number. If we plug something a little greater than $-c$ into f' , we'll get a positive number.



The derivative of f , f' , does not change sign on the intervals $(-\infty, -c)$ and on $(-c, \infty)$, therefore, we have the above chart. From the chart, we can conclude that f has a local minimum at $x = -c$, and that f also has a global minimum there. We could also have used our understanding of the family of functions $g(x) = x^4$ to reason $x = -c$ would be a minimum.

At $x = -c$, f has a local minimum by the first derivative test since $x = -c$ is a critical point of f and the sign of f' changes from negative to positive around $x = -c$.